



LUND
UNIVERSITY

Written Examination
Group and Ring Theory
Monday August 16, 2021
Duration: 08.00–13.00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Solutions may of course be presented in Swedish! Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

The solutions shall be uploaded in the form of a single pdf-file in the Canvas system no later than 13:15. No collaboration or books are allowed. Questions concerning the formulation of the problems can be emailed to Arne Meurman in the Canvas system, who will check the mailbox at 09:30, 11:00. The computer may only be used to read the problems, upload the solutions, and possibly email questions to Arne Meurman.

1. Consider the additive group $(\mathbb{Q}, +)$ as a $\mathbb{Z}$ -module. Show that if

$$f : \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \rightarrow \mathbb{Q}$$

is a group homomorphism, then there exist $c \in \mathbb{Q}$ such that $f(q \otimes r) = cqr$ for all $q, r \in \mathbb{Q}$.

2. Let G be a finite group and let N be a normal subgroup of G . Show that if P is a Sylow- p subgroup of G for some prime p , then $N \cap P$ is a Sylow- p subgroup of N . Show also that NP/N is a Sylow- p subgroup of G/N .

3. Determine the rational canonical form of

$$A = \begin{bmatrix} 1 & 3 & 2 \\ -1 & 5 & 2 \\ 1 & -3 & 0 \end{bmatrix},$$

considered as a complex matrix.

4. Give an example of a finite group G having three Sylow- p subgroups P, Q, R , for some prime p , such that $P \cap Q = \{e\}$ and $P \cap R \neq \{e\}$.
5. Let R be a ring with unity, and let M be an artinian R -module. Let

$$f : M \rightarrow M$$

be an injective R -homomorphism. Show that f is surjective.

6. Let R be a commutative ring with unity 1 such that R has a subring k , which is a field and contains 1 (i.e. R is a k -algebra). Assume that $\dim_k R < \infty$. If R has no nonzero nilpotent elements, show that R is completely reducible as a module over itself.